

SMARTBRIDGE

Let's Bridge the Gap | SKILL WALLET - A SmartBridge Product

Date	29-09-2026
Team ID	SWTID-2026-7949
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Data Flow Diagram (DFD)

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or external actor that sends or receives data (e.g., EEE User, System Administrator).
Rectangle with numbered header	Process	An activity or logic engine that transforms incoming data into outgoing data (e.g., Role Verification, Script-Controlled ACL Rule Engine).
Rectangle (solid fill, no number)	Data Store	A repository where data is held for later access (e.g., User & Role Store [sys_user, sys_user_has_role], Institution Details Table [u_institution_details]).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores (e.g., Read Query, Branch Condition Check, ACL Allow/Deny Signal).

Project Architecture: Script-Controlled ACL Data Flow Specification

DFD Component	Type	Inputs / Incoming Flow	Processing Logic / Function	Outputs / Outgoing Flow
EEE User / Admin	External Entity	User credentials, CRUD request	Initiates record access / submission via ServiceNow UI or API	HTTP Record Requests / Action triggers
1.0 Session & Role Evaluator	Process	Login session, User sys_id	Queries assigned roles (admin, bb1, bb2, bb3, bb4) using gs.hasRole()	Active security role context
D1: User & Role Store	Data Store	Role mappings from admin	Maintains sys_user, sys_user_role, and sys_user_has_role database tables	User authorization parameters
2.0 Script-Controlled ACL Rule Engine	Process	Role context, Target Record (u_institution_details)	Evaluates 3-tier ACL pipeline: (1) Role check, (2) Branch == 'EEE' condition, (3) JavaScript execution logic	Access Grant (true) or Access Deny (false)
D2: Institution Details Store	Data Store	CRUD database instructions	Stores student roll number, name, faculty, branch	Persisted student records

			(ECE/EEE/CSE), email, and phone in u_institution_details	
3.0 Record Rendering & Security Output	Process	Authorized record rows / Deny flag	Filters list/form UI elements, generates 'No records to display' for unauthorized requests	Filtered UI list view, form edit rights, or action errors

Data Flow Architecture Pipeline:

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[External Entity: User] --(1. Request: Read/Create/Write/Delete)--> [Process 1.0: Role Validator] <-
-(Query Roles)-- [Data Store D1: sys_user_has_role]
|
(User Role Context: admin / bb1..bb4)
v
[Process 2.0: Script-Controlled ACL Rule Engine]
|-- Evaluates: (a) Role Requirement: bb1/bb2/bb3/bb4
|-- Evaluates: (b) Field Value Condition: current.branch == 'EEE'
|-- Evaluates: (c) Advanced Script: if (gs.hasRole('admin') || gs.hasRole('bb1'))
return true;
|
+-----+-----+
| [ALLOW] | [DENY] |
v v
[Data Store D2: u_institution_details] [Display Security Access Denied]
| |
+-----+-----+
v
[External Entity: User List / Form Interface View]

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